Volume of Cones, Pyramids and Spheres

Name _____ **Date** _____ Calculator allowed Version 1

KEY WORDS cone · pyramid · sphere · hemisphere · apex · one third · cube root

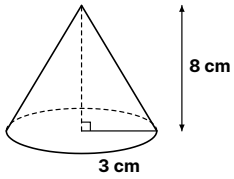
A Cones ●○○

ANSWER ON THE SHEET

Remember: A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

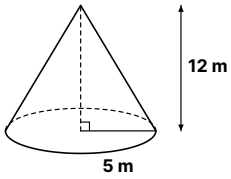
1. Find the volume of each cone.

a)



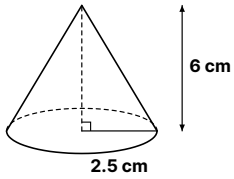
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b)



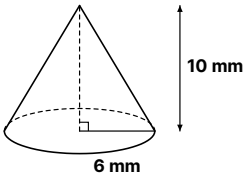
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c)



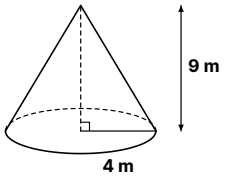
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d)



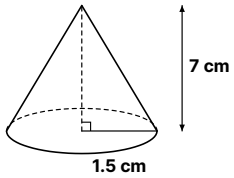
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e)



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f)



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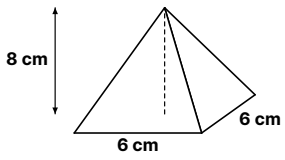
B Pyramids ●●○

ANSWER ON THE SHEET

Remember: A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

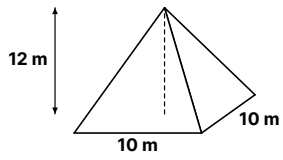
2. Find the volume of each pyramid.

a)



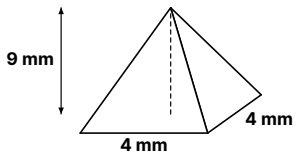
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b)



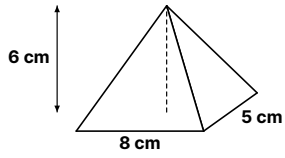
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c)



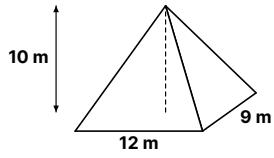
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d)



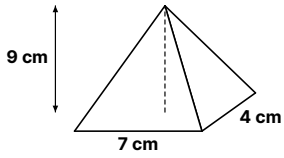
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e)



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f)



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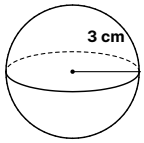
C Spheres and hemispheres ●●○

WORK IN YOUR BOOK

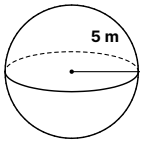
Remember: Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

3. Find the volume of each solid, to 1 decimal place.

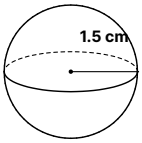
a)



b)



c)



d) A hemisphere (half a sphere) with radius 10 mm

e) A hemisphere (half a sphere) with radius 4 cm

f) A hemisphere (half a sphere) with radius 2.5 m

D Working backwards ●●●

WORK IN YOUR BOOK

Remember: Put the volume into the formula and solve for the missing length. For a sphere you end with r^3: take the cube root.

4. Find the missing length.

a) A cone has a volume of 150 cm³ and a radius of 4 cm. Find its height, to 1 decimal place.

b) A square-based pyramid has a volume of 400 m³ and base edges of 10 m. Find its height.

c) A sphere has a volume of 500 cm³. Find its radius, to 1 decimal place.

d) A cone has a volume of 1,200 mm³ and a radius of 9 mm. Find its height, to 1 decimal place.

e) A square-based pyramid has a volume of 96 cm³ and base edges of 6 cm. Find its height.

f) A sphere has a volume of 3,000 mm³. Find its radius, to 1 decimal place.

E Problems ●●●

WORK IN YOUR BOOK

5. An ice cream is a cone of radius 3 cm and height 10 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

6. Sand is tipped into a conical pile with a radius of 2.5 m and a height of 1.8 m. Find the volume of sand, to 2 decimal places.

7. A ball of diameter 20 cm is packed in a cube-shaped box with edges of 20 cm. Find the volume of empty space in the box, to the nearest cm³.

8. A roof is a square-based pyramid with base edges of 8 m and a height of 3 m. Find the volume of the roof space.

★ Extension

FOR EARLY FINISHERS

9. A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

10. A sphere of radius 6 cm is melted and recast as a cone with the same radius. Find the height of the cone.

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Volume of Cones, Pyramids and Spheres

Name _____ **Date** _____ Calculator allowed **Version 2**

KEY WORDS cone · pyramid · sphere · hemisphere · apex · one third · cube root

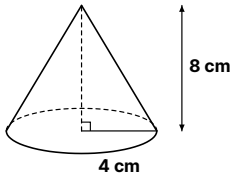
A Cones ●○○

ANSWER ON THE SHEET

Remember: A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

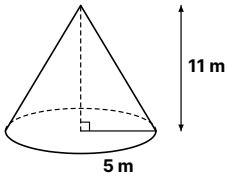
1. Find the volume of each cone.

a)



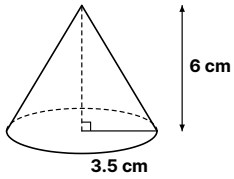
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b)



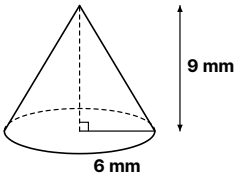
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c)



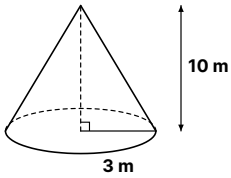
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d)



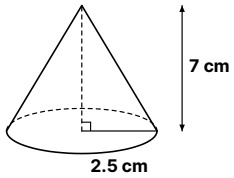
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e)



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f)



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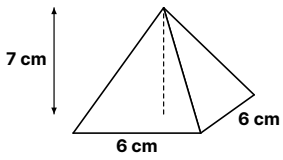
B Pyramids ●●○

ANSWER ON THE SHEET

Remember: A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

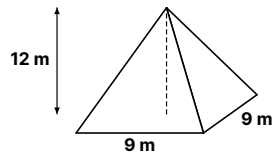
2. Find the volume of each pyramid.

a)



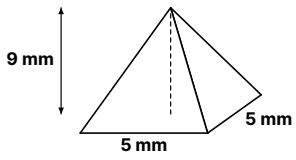
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b)



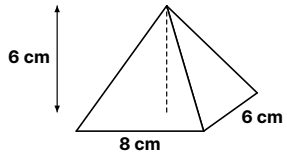
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c)



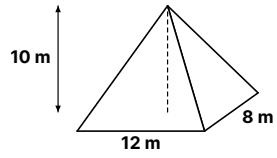
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d)



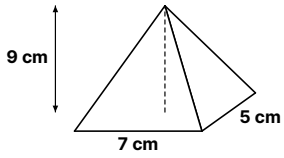
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e)



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f)



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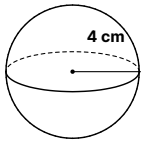
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C Spheres and hemispheres ●●○ WORK IN YOUR BOOK

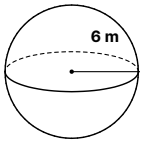
Remember: Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

3. Find the volume of each solid, to 1 decimal place.

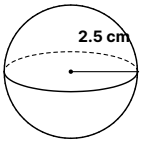
a)



b)



c)



d) A hemisphere (half a sphere) with radius 9 mm

e) A hemisphere (half a sphere) with radius 5 cm

f) A hemisphere (half a sphere) with radius 3.5 m

D Working backwards ●●● WORK IN YOUR BOOK

Remember: Put the volume into the formula and solve for the missing length. For a sphere you end with r^3: take the cube root.

4. Find the missing length.

a) A cone has a volume of 200 cm³ and a radius of 5 cm. Find its height, to 1 decimal place.

b) A square-based pyramid has a volume of 324 m³ and base edges of 9 m. Find its height.

c) A sphere has a volume of 800 cm³. Find its radius, to 1 decimal place.

d) A cone has a volume of 1,500 mm³ and a radius of 10 mm. Find its height, to 1 decimal place.

e) A square-based pyramid has a volume of 128 cm³ and base edges of 8 cm. Find its height.

f) A sphere has a volume of 4,000 mm³. Find its radius, to 1 decimal place.

E Problems ●●● WORK IN YOUR BOOK

5. An ice cream is a cone of radius 3.5 cm and height 11 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

6. Sand is tipped into a conical pile with a radius of 3 m and a height of 2.1 m. Find the volume of sand, to 2 decimal places.

7. A ball of diameter 24 cm is packed in a cube-shaped box with edges of 24 cm. Find the volume of empty space in the box, to the nearest cm³.

8. A roof is a square-based pyramid with base edges of 9 m and a height of 4 m. Find the volume of the roof space.

★ Extension FOR EARLY FINISHERS

9. A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

10. A sphere of radius 9 cm is melted and recast as a cone with the same radius. Find the height of the cone.

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